

Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ECE 2127/EC_ 2127 DIGITAL CIRCUITS
(SPL: ELECTRONICS ENGINEERING (VLSI DESIGN AND TECHNOLOGY))

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a combinational circuit using 4:16 decoder with active-low outputs to convert BCD code to equivalent XS-3 code. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1B) Design a combinational circuit using programmable logic array (PLA) to implement a full subtractor. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 1C) Use Boolean algebra to derive the simplest sum-of-products expression for the function (3)
 $f = x\bar{y}z + wyz + w\bar{x}z$. Implement the simplified function using only NAND gates.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Design a Moore-type circuit using D flip-flops, with one input and one output to detect the occurrences of the bit pattern 110 in a sequential pulse train. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 2B) Implement the function $f(w, x, y, z) = \sum m(0, 2, 4, 5, 7, 9, 10, 14)$ using 2-level multiplexer structure comprising of two 4:1 MUXs and one 2:1 MUX. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Using the characteristic equations and excitation table, realize an SR flip-flop using a D flip-flop. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 3A) Design a hazard-free mod-16 asynchronous edge counter by synthesizing a mod-8 pulse counter. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Using negative edge triggered T flipflops, design mod 14 ripple counter. Also draw the timing diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 3C) Analyze the behavior of a SR flip-flop as an asynchronous sequential circuit using the Moore model. Derive the state transition table, flow table and state diagram accordingly. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4A) Design a synchronous Mod-12 counter using JK flipflops. Draw the timing diagram for the same. (4)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4B) Using a universal shift register design a circuit to generate the following sequence: 1001110100111010... (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 4C) Design a sequential circuit to generate the following sequence 101110100111100101110.. using indirect logic. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5A) Design an optimally state-assigned Mealy synchronous sequential circuit for serial binary adder using JK flip-flops and logic gates. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Plot the timing diagram showing the Q and $\bar{Q}$ outputs of Master-slave SR flipflop, with the control input C, based on the given waveform. (3)



Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Design a synchronous sequential circuit using edge-triggered JK flip-flops to generate the sequence: 0 → 2 → 4 → 3 → 5 → 6 → 0 → 2 → ... Ensure that any invalid or undesired states are redirected to state 3. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2127:Digital Circuits

#	Course Outcome	Description	Marks
1	1	CO 1	3
2	3	CO 3	10
3	4	CO 4	15
4	5	CO 5	22
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2127:Digital Circuits

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	15	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)